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SUPERSEDING
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MILITARY SPECIFICATION
CONTROL, INTERCOMMUNICATION SET C-10456( )/VRC

This specification is approved for use by Communications Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for Control, Intercommunication Set C-10456( )/VRC, which is intended for use in vehicular communication systems. (See 6.1 and 6.8).

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

Federal

NN-P-71	Pallet, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
PPP-B-20	Bag, Cotton, Mailing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications Research and Development Command, ATTN: DRDCO-CM-DM, Fort Monmouth, New Jersey 07703, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC-5820

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PPP-T-97	Tape, Pressure Sensitive Adhesive, Filament Reinforced
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-T-495	Tube, Mailing and Filling
PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed and Lock Corner
PPP-B-636	Box, Shipping, Fiberboard
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall
PPP-S-760	Strapping, Nonmetallic, (and Connectors)
PPP-C-843	Cushioning, Material, Cellulosic
QQ-S-781	Strapping, Steel, Flat and Seals

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-S-901	Shockproof Equipment, Class HI (High Impact), Shipload
MIL-I-6869	Impregnants for Aluminum and Magnesium
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Insepction by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads. 40 inch by 48 inch Pallets, Skids, Runners or Pallet Type Base
MIL-STD-252	Wire Equipment, Classification of Visual and Mechanical Defect
MIL-STD-276	Impregnation of Porous Nonferrous Metal Castings
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-965	Parts Control Program
MIL-STD-1188	Commercial Packaging of Supplies and Equipment

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DRAWINGS

Communications R&D Command

DLB4001090	Control, Intercommunication Set C-10456( )/VRC

(Copies of specifications, standards, and drawings required by the contractor in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Construction. The equipment shall be constructed in accordance with drawing and Data List DLB4001090 and the requirements of this specification.

3.2 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein, shall apply, except that MIL-STD-965 shall be used in lieu of MIL-STD-749 for parts control procedures.

3.4 Finish, protective. Equipment shall be given protective finish in accordance with Specification MIL-F-14072. This includes finish of hardware, such as handles, hinges, screws, etc, and necessary touch up after mounting. The final paint film on Type I surfaces shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency. (See 6.3).

3.5 Marking.

3.5.1 General. Marking shall conform to Specification MIL-M-13231. (See 4.4). Front panel marking shall be group I as described in that specification.

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3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. Control, Intercommunication Set C-10456( )/VRC shall be serial numbered.

3.6 Die castings. (See 4.4).

3.6.1 Quality of die castings. Quality of die castings shall be as follows:

a. Surface condition shall be clear, dry and free from chips.

b. Cast surface finish shall be of a painting grade which allows some streaks and chill marks which can be covered and hidden by paint.

c. Flash must be removed by grinding or filing.

d. The casting, after impregnation, shall be air tight, and no leakage shall be allowed under a pressure of 10 psi gage.

e. Flatness shall be as specified in the government furnished drawings.

f. Dimensions shall be as specified in the government furnished drawings.

3.6.2 Impregnation of aluminum die castings. All aluminum sand and die castings shall be impregnated in accordance with Standard MIL-STD-276 using impregnating material Type II per Specification MIL-I-6869. Castings which have been impregnated shall be marked "IMP". The impregnation shall be to correct general seepage only and shall not be used to correct poor foundry techniques or significant porosity (see 4.4).

3.7 Service conditions. The equipment shall meet the following service conditions:

3.7.1 Temperature. The equipment shall be subjected to the temperature procedure as shown in Method 501.1 and 502.1 of MIL-STD-810. Measurements shall be taken with the equipment operating with 22-volt input and with 30-volt input.

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a. Low. Both during and after testing as specified in 4.10.1.1, the equipment shall meet the requirements of para. 3.9.3 (audio output power) and para. 3.9.7 (distortion).

b. High. Both during and after testing as specified in 4.10.1.2, the equipment shall meet the requirements of para. 3.9.3 (audio output power) and para. 3.9.7 (distortion).

3.7.2 Humidity (Moisture-Resistance). Both during and after testing as specified in 4.10.2, the equipment shall meet the requirements of para. 3.9.3 (audio output power) and para. 3.9.7 (distortion).

3.7.3 Altitude (Elevation). Both during and after testing as specified in 4.10.3, the equipment shall meet the requirements of para. 3.9.3 (audio output power) and para. 3.9.7 (distortion).

3.7.4 Vibration and shock.

3.7.4.1 Vibration, internal (See 4.10.4). The equipment shall withstand vibration associated with vehicular transport as tested in accordance with para. 4.10.4. There shall be no physical damage and the equipment shall meet the requirements of para 4.14.

3.7.4.2 Shock. The equipment shall be capable of meeting the requirements of Table I.

Table I - Shock

Paragraph Heading	Paragraph	Performance After Test
Shock, basic design	4.10.5	Specified performance (Note A). No physical damage.
Shock, ballistic	4.10.6	Operable (Note A). Any physical damage shall be minor only, as classified in MIL-STD-252.

NOTE A: The equipment shall meet the requirements of paragraph 4.14.

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3.7.5 Immersion. The equipment shall show no evidence of leakage after it is tested as specified in 4.10.7.

3.8 Preconditioning. (See 4.6).

3.8.1 Bounce preconditioning. The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1.

3.8.2 Burn-in (debugging). Each Control, Intercommunication Set on order shall be subjected to a burn-in operation of 24 hours at 25.5 volts dc input. The last six hours of the 24 hour burn-in shall be failure free. Control, Intercommunication Set shall meet the requirements specified herein, without subsequent processing after subjection to paragraph 4.6.2.

3.9 Electrical requirements.

3.9.1 Input voltage range. The control unit shall be capable of meeting full specification requirements when operating from a 22 to 30 Vdc source (see 4.9.6).

3.9.2 Microphone amplifier output circuit. This circuit shall have an output impedance of 1200 ohms minimum when measured in accordance with 4.9.1.

3.9.3 Microphone amplifier audio output power. The output level shall be .22 V + 2 dB, - 4 dB, into a 150 ohm load when measured in accordance with 4.9.2.

3.9.4 Microphone amplifier input circuit. This circuit shall have an impedance of 150 ohm $\pm$ 15% when measured in accordance with 4.9.7.

3.9.5 Volume control. The volume control located on the bottom of the control unit shall vary the headphone output level. At minimum setting of the control, there shall not be more that .02 mw at the headphone output. At maximum setting of the control, the output levels shall be a minimum of 90 mw when tested in accordance with 4.9.3.

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3.9.6 Audio response. The audio response of the microphone amplifier shall be within ± 2 dB of the 1000 Hz reference from 500 to 3000 Hz when tested in accordance with 4.9.4.

3.9.7 Distortion. When measured in accordance with 4.9.5 the overall total harmonic distortion shall not exceed 2 percent.

3.9.8 Control function. The C-10456( )/VRC control unit shall provide the following functions:

3.9.8.1 Selection of monitoring function. Test in accordance with 4.9.8.

3.9.8.2 Selection of transmission function. Test in accordance with 4.9.9.

3.9.8.3 Remote interphone/radio keying function. Test in accordance with 4.9.10.

3.9.8.4 Provide a pathway for the line function between the two 18-pin receptacles, and between specified pins of two audio receptacles. Test in accordance with 4.9.11.

3.10 Operational inspection. The units shall be given a final inspection, as specified in 4.14, after all other inspections and tests are completed, to determine that they are operable prior to packaging.

3.11 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454. (See 4.11)

3.12 Air seal. The equipment case shall provide means for enabling performance of the air-seal test. When tested as specified in 4.7, the decrease in vacuum shall not exceed 0.01 pounds per square inch, gauge, per minute.

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3.13 Technical literature and running spare parts. Technical literature and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2).

3.14 Plug-in subassemblies. The control units shall contain a plug-in microphone amplifier assembly.

3.15 Fungus. Equipment and assemblies shall show no evidence of viable fungus or corrosion* when subjected to 28 days of fungus test in accordance with paragraph 4.10.9

* Corrosion is defined as any visible degradations that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.16 Salt fog. The equipment and assemblies shall be exposed, in their operating configuration, to 48 hours of salt fog in accordance with paragraph 4.10.8, and shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered or otherwise loosened finish or metal surfaces. There shall also be no impairment of operation of any control feature of the equipment.

3.17 Safety engineering. (See 4.12).

3.17.1 Personnel hazard. Personnel hazards shall be kept to a minimum through compliance with Requirement 1 of MIL-STD-454.

3.17.2 Edge rounding. All exposed edges and corners shall be rounded sufficiently to insure personnel safety.

3.17.3 Radioactive materials. Radioactive materials shall not be used.

3.18 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

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a. In MIL-P-11268:

General requirements for plastic material and plastic parts.	Cleaning of units
Wiring and cabling, including:	General requirements for securing of parts.
Slack.	
Protection.	
Clearance.	
Splicing.	Controls.
Connections, general.	Self-tapping screws.
Grounding, general.	
Shielding on wire and cable.	
Soldering and brazing.	

b. In this specification:

3.4	Finish, protective
3.5	Marking
3.6	Die castings

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First Article Inspection (does not include packaging). (See 4.3).

b. Inspection covered by subsidiary documents. (See 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging. (See 4.5).

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(2) Quality conformance inspection of packaging. (See 4.15).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish four (4) first article units of the complete Control, Intercommunication Set C-10456( )/VRC.

4.3.2 First article inspection. The first article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning specified in 4.6, and the inspections specified for Group A, Group B and Group C (see Tables III, IV and V respectively). The inspection shall be performed in the following order, 4.4, 4.6, Group A and Group B for all units, and Group C as specified in Table II. After completion of the Group C environmental tests, conforming units shall be reinspected and shall pass all Group A inspection.

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Table II - Order Of Environmental Tests

Inspection	NUMBER OF SAMPLES			
	Unit 1	Unit 2	Unit 3	Unit 4
Non-damaging				
Temperature				
Low				1
High				2
Altitude (elevation)			1	
Potentially damaging				
Salt Fog				3
Vibration (internal)	1			
Shock, basic design	2			
Shock, Ballistic	3			
Humidity (Moisture Resistance)		1		
Fungus			2	
Immersion	4			

NOTE 1: Other First Article inspection may precede, follow or be interspersed between the non-damaging tests.

NOTE 2: The order on non-damaging tests and the choice of which units are subjected to the non-damaging tests may be varied if convenient, except that the vibration test should be performed on the same unit used for the shock test.

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4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where Required
Parts, Materials and Processes Used In Electronic Equipment	3.3
Finish	3.4
Marking	3.5
Casting, Impregnation and Quality	3.6

4.5 Quality conformance inspection of equipment before packaging. Each unit of each lot of equipment produced under this specification shall be subjected to bounce preconditioning (see 3.8.1) and then to a 24 hour burn-in operation (see 3.8.2) prior to selection of samples for inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the government -- at its discretion -- may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.9). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table III. Each lot will be subject to sampling inspection, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQLs indicated in Table III. Group A Inspection shall be performed in any order which is satisfactory to the government except that the airseal test (4.7) shall be next to last and the operational inspection (4.14) shall be last. (See 6.7).

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Table III - Group A Inspection

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical				
Mike Amplifier, A-80	3.18	4.13	0.65%	2.5%
Control with Cover off	3.18	4.13	1.0%	4%
Control C-10456( )/VRC	3.18	4.13	1.0%	4%
Electrical				
Module A-80 Control Box C-10456( )/VRC	SC-A-324072		0.65%	N/A
Mike Amplifier Output Impedance	3.9.2	4.9.1		
Audio Output	3.9.3	4.9.2		
Mike Amplifier Input Impedance	3.9.4	4.9.7	1% for the Group	N/A
Volume Control	3.9.5	4.9.3		
Audio Response	3.9.6	4.9.4		
Distortion	3.9.7	4.9.5		
Air Seal	3.12	4.7	1%	N/A
Operation	3.10	4.14	1%	N/A

4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table IV and to the special procedure for small-sample inspection of Standard MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and passed the Group A inspection. Inspection Level shall be S-4.

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Table IV - Group B Inspection

Inspection	Req Para	Insp Para	AQL
Control Functions			
Selection of Monitoring functions	3.9.8.1	4.9.8	
Selection of Transmission functions	3.9.8.2	4.9.9	4% for the group
Remote Keying	3.9.8.3	4.9.10	
Line Function	3.9.8.4	4.9.11	
Input Voltage Range	3.9.1	4.9.6	
Immersion	3.7.5	4.10.7	6.5%
Interchangeability	3.11	4.11	6.5%

4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the government.

4.5.3 Group C inspection. This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and passed Group A and Group B inspection. Sample units shall be selected in accordance with paragraph 4.5.3.1.

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Table V - Group C Inspection

Inspection	Req Para	Insp Para
Subgroup I		
Temperature	3.7.1	4.10.1
Humidity	3.7.2	4.10.2
Elevation	3.7.3	4.10.3
Vibration, internal	3.7.4.1	4.10.4
Shock, ballistic	3.7.4.2	4.10.6
Shock, basic design	3.7.4.2	4.10.5
Subgroup II		
* Salt Fog	3.16	4.10.8
* Fungus	3.15	4.10.9

*Unless otherwise specified in contract or order, a different sample unit shall be used for each of these tests. Following fungus and salt fog testing, the test item shall be thoroughly washed, cleaned, dried, and refurbished, if necessary before proceeding with subsequent tests.

4.5.3.1 Sampling for inspection of equipment. Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.3.1.1 Subgroup I. For this subgroup, two units shall be selected from the first 50 production units for the first Group C inspection. For subsequent Group C inspections, two units every 6 months shall be selected.

4.5.3.1.2 Subgroup II. For this subgroup, one unit shall be selected from the first production lot for the first Group C inspection. For subsequent Group C inspections, two units shall be selected every 9 months.

4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the government.

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4.5.3.3 Noncompliance. Actions relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group B and Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.5.5 Disposition of Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order.

4.6 Preconditioning. (See 3.8).

4.6.1 Bounce preconditioning. The unit, with shock mounts (if any) removed or blocked, shall be placed on its cover on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneatales, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. Each unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned or otherwise changed prior to subjection to inspection. (See 3.8.1).

4.6.2 Burn-in debugging. (See 3.8.2).

4.6.2.1 Test conditions. Each Control, Intercommunication Set on order shall be subjected to a burn-in period of 24 hours with an input signal of 7.0 mV rms, at a frequency of 1000 Hz; the microphome amplifier shall deliver .22 Vrms + 2 dB, - 4 dB, to a 150 ohm load at a temperature of 40°C ± 5°C (95°F to 113°F); while the unit is in this operating mode, vibration shall be performed as follows:

2.2G ± 10% at any non-resonant frequency between 20 and 60 Hz measured at the mounting points of the equipment. This vibration shall be applied for 10 minutes during each hour of equipment operating time.

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4.6.2.2 Failures. In the event of a failure during burn-in debugging, the contractor will perform all necessary rework (including replacement of components, modules and wiring, etc.) to correct the failure cause. If the failure occurs during the first 18 hours of the burn-in period, the unit will be reworked and returned to burn-in. If the failure occurs during the last 6 hours of the burn-in period, the unit will be reworked and resubmitted to burn-in until it successfully survives the 6 hour failure free period. If after successful completion of burn-in debugging, the equipment fails either the Group A or Group B tests, the equipment will be reworked and resubmitted to burn-in debugging.

4.6.2.3 Compliance. Units passing burn-in debugging will be submitted to Group A, Group B and Group C inspection, in that order, without benefit of repair, cleaning, realignment, or other changes, unless otherwise specified herein.

4.7 Air-seal test. The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment inclosure. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed 0.01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily. A screw plug shall be provided per government drawings for the air-seal test. (See 3.12).

4.8 Standard test conditions. All tests shall be conducted under the following conditions, except as specified for the individuals tests.

Temperature	Room ambient
Humidity	Room ambient
Input voltage	25.5 volts dc
Input voltage, audio	7.0 mV rms
Audio load, impedance	600 ohms
Audio frequency	1000 Hz
Source impedance	150 ohms
Microphone amplifier load impedance	150 ohms
Shorting plug P805 connected	
Pin C of J802 or J803 grounded	

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4.9 Electrical tests.

4.9.1 Microphone amplifier output impedance. Under standard test conditions except no load, measure the output voltage. Terminate the output with a 1200 ohm resistor and measure the output voltage. The open circuit voltage shall be at least twice the loaded value (see 3.9.2).

4.9.2 Microphone amplifier audio output power. Under standard test conditions measure the output voltage for compliance with 3.9.3.

4.9.3 Volume control. The control unit shall be operated under standard test conditions. The headphone output shall be measured for full clockwise rotation and counterclockwise settings of the volume control with an input voltage of 7.75 volts rms to establish compliance with 3.9.5.

4.9.4 Audio response. The control unit shall be operated under standard test conditions. The output voltage shall be measured, then the frequency of the input signal shall be changed to 500, 2000 and 3000 Hz. The output shall conform to 3.9.6.

4.9.5 Distortion test. The control unit shall be operated under standard test conditions and shall comply with 3.9.7.

4.9.6 Input voltage range tests. To determine compliance with 3.9.1, the tests of 4.9.2 and 4.9.5 shall be performed at input voltages of 22 volts dc and 30 volts dc.

4.9.7 Input impedance test. Input impedance shall be measured at 1000 Hz to establish compliance with 3.9.4.

4.9.8 Monitoring functions. Turn volume control to maximum clockwise, apply standard load and connect 7.75 volts 1000 Hz to pin M (unmuted audio from "A" radio) of J804. Measure 7.35 volts minimum across 600 ohm load (connected to pin E of either J802 or J803) when selector switch is placed in A position. Repeat above for other selector switch positions as follows: (See 3.9.8.1).

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Apply audio to pin B (unmuted audio from "B" radio) of J804, selector in B position.

Apply audio to pin J (unmuted audio from "C" radio) of J804, selector in C position.

Apply audio to pin L (fixed audio) of J804, selector in ALL position.

Apply audio to pin E (fixed audio) of J804, selector in INT ONLY position.

4.9.9 Transmission control. Remove ground from pin C of J802 and J803 and ground only as required for test.

a. Control of "A" radio. Apply 25.5 volts dc to pin D (keying control "A" radio) of J804 through a 1000 ohm resistor. Ground pin C (radio key) of J802. The voltage at pin D of J804 shall be less than 1 volt in selector positions ALL, A and B, and shall be at least 24 volts in selector positions C and INT ONLY.

b. Control of "C" radio. Apply 25.5 volts dc to pin F (keying control to "C" radio) of J804 through a 1000 ohm resistor. Ground pin C of J802. The voltage at pin F of J804 shall be less than 1 volt in selector position C and more than 24 volts in all other selector positions.

c. Control of intercom. Apply 25.5 volts dc to pin H (intercom keying) of J804 through a 1000 ohm resistor. Place monitor switch in ALL position and ground pin C of connector J803. The voltage at pin H of J804 shall be less than 1 volt. Place monitor switch in INT ONLY position. Grounding pin C of either J802 or J803 shall reduce the voltage at pin H of J804 to less than 1 volt. (See 3.9.8.2).

4.9.10 Remote keying function. Remove shorting plug P805 from J805 and remove ground from pin C of J802 and J803. Apply 25.5 volts dc to pin D of J804 through a 1000 ohm resistor. Put monitor switch in position A. Ground pin C of J803. The dc voltage at pin D of J804 shall be at least 24 volts, and the audio output at pin K of J804 shall be less than 1 millivolt rms. Connect a jumper between pins 1 and 3 of J805; the voltage at pin D of J804 shall decrease to less than 1 volt and the audio output at pin K of J804 shall be 0.22 volts + 2 dB, - 4 dB. (See 3.9.8.3).

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4.9.11 Line pathway. Test for continuity between pins A, B, C, D, E, F, H, J, K, L, M, N, U and V of J804 and corresponding pins of J801. Test for continuity between pins A, B, D and E of J802 and corresponding pins of J803. Test for continuity between pins B and E of J802. (See 3.9.8.4).

4.10 Environmental service conditions tests.

4.10.1 Temperature test.

4.10.1.1 Low temperature. The equipment shall be tested in accordance with Method 502.1, Procedure I, of MIL-STD-810. The storage temperature shall be -80°F, and shall be maintained until equipment stabilization, and the operating temperature shall be -40°. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.7.1a.

4.10.1.2 High temperature. The equipment shall be tested in accordance with Method 501.1, Procedure II, of MIL-STD-810. The operating temperature shall be +150°F. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.7.1b.

4.10.2 Humidity (moisture-resistance) test. The equipment shall be tested in accordance with Method 507.1, Procedure II of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.7.2. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test. The test shall be performed with the cover off.

4.10.3 Altitude (elevation) test. The equipment shall be tested in accordance with Method 500.1, Procedure I, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.7.3.

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4.10.4 Vibration. The equipment shall be tested in accordance with Method 514.2, Procedure VIII, using curve W, of MIL-STD-810. (See 3.7.4.1).

4.10.5 Shock, basic design. The equipment shall be tested in accordance with method 516.2, Procedure I, figure 516.2-1, parameters b and d of MIL-STD-810. (3.7.4.2).

4.10.6 Shock test; ballistic.

a. Test method. The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in Specification MIL-S-901. The control unit shall be secured in its normal operating position to test plate fixture 4A, figure 5 of MIL-S-901, by means of the same fasteners used for vehicular installation of the control. The test shall consist of a total of 9 blows: one each 1 foot blow, 3 foot blow, and 5 foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the control under test is permissible. (See 3.7.4.2).

b. Performance. Upon completion of the test and before repairs are made, tests shall be made for compliance with the requirements of para 4.14 (operation inspection).

4.10.7 Immersion test (See 3.7.5). The equipment, as prepared for field transportation, shall be tested in accordance with Method 512.1, Procedure I, of MIL-STD-810.

4.10.8 Salt Fog. The equipment and assemblies shall be exposed in their operating configuration to salt fog in accordance with Method 509.1, Procedure I of MIL-STD-810 for 48 hours. (See 3.16).

4.10.9 Fungus. The equipment and assemblies shall be subjected to 28 days of the fungus test specified in Method 508.1 of MIL-STD-810. For gasket or other non-hermetically sealed equipments, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The equipment shall be subjected to the test in this open condition, for compliance with paragraph 3.15.

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4.11 Inspection for interchangeability. The points listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.11. When listed dimension is not within specified or design limits, it shall be considered a major defect.

a. Case. The outside dimensions; location of four cover mounting screw holes, and cover alignment pins.

b. Bottom cover. The maximum overall outside dimension, length and width, and location of four mounting screws and three cover alignment holes.

4.11.1 Electrical performance of A-80 module. Prior to assembly in the control, the A-80 module shall be tested to determine compliance with the performance requirements specified in SC-A-324072.

4.12 Safety engineering. An inspection shall be performed to verify compliance with those portions of 3.17 which can be determined visually.

4.13 Visual and mechanical inspection. The equipment shall be examined for defects listed in Standard MIL-STD-252.

4.14 Operation inspection. The control shall be operated under the standard test conditions specified in 4.8 and shall comply with the requirements for the microphone amplifier audio output power (see 3.9.3), distortion (see 3.9.7), and control functions (see 3.9.8). (See 3.10).

4.15 Quality conformance inspection. Quality conformance inspection of Section 5 requirements shall be in accordance with the Group A, Group B, and, when required, (see 6.2), the Group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

5. PACKAGING

5.1 Preservation. Preservation shall be level A, B or Commercial, as specified (see 6.2).

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5.1.1 Level A.

5.1.1.1 Cleaning. Each Control, Intercommunication Set C-10456( )/VRC shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Control, Intercommunication Set C-10456( )/VRC shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Hardware. Each set of hardware shall be stowed in a cotton mailing bag conforming to PPP-B-20 and the bag opening secured with the drawstrings.

5.1.1.4.2 Control, Intercommunication Set C-10456( )/VRC. Each control shall be unit packed method III as follows: When required, place the dust covers on the connectors. Disconnect shorting plug P805 from J805 and cushion it with cushioning conforming to PPP-C-843, type II, class B, 1/4 inch thick, position it on top (or side) of control and secure with tape conforming to PPP-T-76. Position a close-fitting fiber tube conforming to PPP-T-495, type I, over and 1/8 inch above the volume control knob. Secure the bag of hardware, unit packed as specified in 5.1.1.4.1, to the monitor knob with drawstrings. Wrap the unit, with plug P805 and bag of hardware attached, with cushioning material conforming to PPP-C-843, type II, class B, 1/2 inch thick. Secure cushioning with tape conforming to PPP-T-76. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Closure shall be in accordance with the appendix of the box specification.

5.1.1.4.3 Unit pack size. Unit pack size shall not exceed .105 cubic feet.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

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5.1.3 Commercial preservation. Preservation shall be in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of controls, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type I, finish A.

5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers, packed as specified in 5.2.1 except that container strapping may be omitted, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV.

5.2.2 Level B. A quantity of controls, unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-640, class 2, style E or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the

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gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to AA-S-781, type I, finish A or shrink film conforming to L-P-378, type IV. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination exceed either a total of 250 pounds (excluding the pallet) or a volume of 20 cubic feet; and the container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of controls, unit packed as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type I, finish A, or shrink film conforming to L-P-378, type IV.

5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. Controls, Intercommunication Sets C-10456( )/VRC are intended for use in tracked vehicles in conjunction with Amplifier, Audio Frequency AM-1780/VRC and with radios of the AN/VRC-12 family. The C-10456( )/VRC is similar to the C-2298( )/VRC but provides for a remote keying capability. With a remote switch (such as a foot switch or special

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hand switch) connected, the crew member does not have to reach up to his combat vehicle crewman helmet switch to key either radio or intercom, but can accomplish these functions with less distraction via the remote switch.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, date of this specification and any amendment thereto.

b. Level A, level B or commercial preservation and packing (see Section 5).

c. First Article pack(s) as follows:

Makeup of pack(s)
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

d. Marking and shipping of samples.

e. Place of final inspection.

f. Technical literature required. (See 3.13).

g. Quantity of running spare parts required. (See 3.13).

h. When rough handling test is required. (See 4.16).

i. Size data. Size limitations cited herein are based on the dimensions of the equipment for which these requirements were established. Requests to exceed these limitations must be supported by data indicating why limitations cannot be met and recommend a revised limitation. (See 5.1.1).

6.3 Color. The color chip X24087 may be obtained upon request to Commanding Officer, US Naval Supply Depot, 5801 Tabor Avenue, Philadelphia, PA 19120.

6.4 Definitions.

6.4.1 Equipment divisions. The terms used herein for equipment divisions conform to Standard MIL-STD-280.

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6.4.2 Definitions of inspection terms. The inspection terms used herein conform to Standard MIL-STD-109. The following are abbreviated definitions selected from those given therein for immediate reference.

6.4.2.1 Inspection. Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.4.2.2 Examination. Examination consists of simple, generally non-destructive determinations of compliance, without use of special testing equipment.

6.4.2.3 Testing. Testing consists of determinations of compliance, using technical means.

6.5 Humidity. Natural climatic extremes may include 100-percent relative humidity, with condensation, up to 4 hours at a time. This requirement (see 3.7.2) is specified by Standard MIL-STD-210. The moisture resistance test (see 4.10.2) using Standard MIL-STD-810, is a procedure that gives a practical test approximating the natural conditions.

6.6 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.2).

6.7 Location of operational inspection and air-seal test. It is desirable that the operational inspection (4.14) and the air-seal test (4.7) be performed at a location that will minimize handling (which might cause damage to the equipment) after these inspections are completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.8 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for

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example: C-10456W/VRC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.9 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.10 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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